Why CUA Failures Are Hard to Diagnose

REAL CASE **Task (abridged): Find unpaid tuition e-mails and add them to the receiver field**

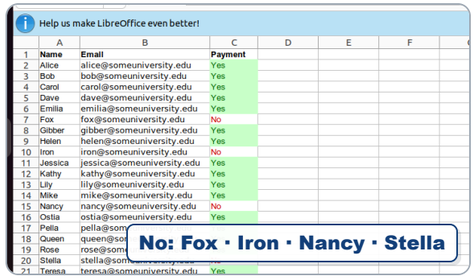
Environment: LibreOffice Calc → Thunderbird

A Failure trajectory

⚠ ROOT CAUSE

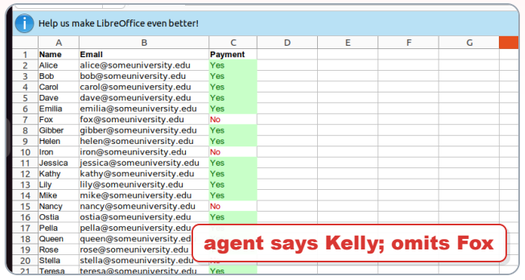
⚠ OBSERVED FAILURE

1 after t=5 Source record



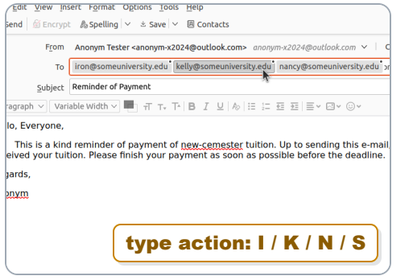
The payment column identifies exactly four unpaid students; row 12 is Kathy with **Yes**.

2 t=6 Root reasoning



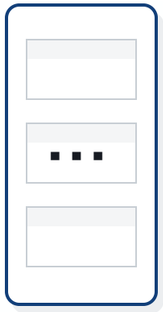
The agent invents **Kelly** at row 12 and drops Fox, even though the visible row is Kathy / Yes.

3 t=10 Recipient entry

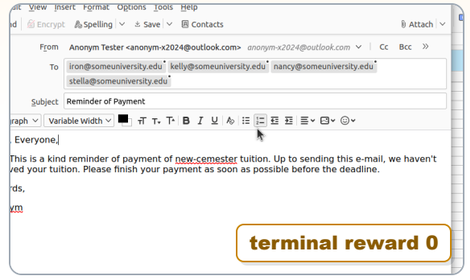


The four-address action carries the false belief into Thunderbird; the complete chips are visible at t=12.

4 1 more action



5 t=12 Final state



The draft still contains Kelly and still omits Fox when the trajectory declares completion.

B Why the root step is subtle

1 Row-level evidence

- 7 — Fox = No
- 12 — Kathy = Yes
- ∅ — No Kelly row

Name, e-mail, and payment status must stay bound by row.

2 Perception to belief

- UI — Visible source: Fox / No; Kathy / Yes — **record**
- R — Agent response: Kelly; no Fox — **P1**

The root is the step-6 reasoning, not the scroll action executed afterward.

3 Intended vs. actual

Human reference

F/I/N/S — Fox / Iron / Nancy / Stella

Agent response · P1

I/K/N/S — Iron / Kelly / Nancy / Stella

C Why the final symptom is insufficient

Expected recipients	Observed at t=12
Fox · Iron · Nancy · Stella	Iron · Kelly · Nancy · Stella

The receiver field proves the list is wrong, but the final frame alone cannot show where Kelly came from.

Hypotheses compatible with the wrong receiver list (which one is correct?)

- ✓ **Hallucinated list at t=6** — RECORDED ROOT · P1
- ○ Source spreadsheet wrong
- ○ Thunderbird entry bug

D Three evidence challenges illustrated by this case

1 Dense table reading

Names, e-mails, and payment states must be joined across rows.

2 Visual hallucination

A plausible name is invented while a visible unpaid row is omitted.

3 Cross-app propagation

The false belief later becomes an incorrect Thunderbird recipient list.